A Multiscale and Multidirection Feature Fusion Network for Road Detection From Satellite Imagery

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***Abstract*— The completeness of road extraction is very important for road application. However, existing deep learning (DP) methods of extraction often generate fragmented results. The prime reason is that DP-based road extraction methods use square kernel convolution, which is challenging to learn long-range contextual relationships of roads. The road often produce fractures in the local interference area. Besides, the quality of extraction results will be subjected to the resolution of remote sensing (RS) image. Generally, an algorithm will produce worse fragmentation when the used data differ from the resolution of the training set. To address these issues, we propose a novel road extraction framework for RS images, named the multiscale and multidirection feature fusion network (MSMDFF-Net). This framework comprises three main components: the multidirectional feature fusion (MDFF) initial block, the multiscale residual (MSR) encoder, and the multidirectional combined fusion (MDCF) decoder. First, according to the road's morphological characteristics, we develop a strip convolution module with a direction parameter (SCM-D). Then, to make the extracted result more complete, four SCM-D with different directions are used to MDFF-initial block and multidirectional combined fusion decoder (MDCF-decoder). Finally, we incorporate an additional branch into the residual network (ResNet) encoding module to build multiscale residual encoder (MSR-encoder) for improving the generalization of the model on different resolution RS image. Extensive experiments on three popular datasets with different resolution (Massachusetts, DeepGlobe, and SpaceNet datasets) show that the proposed MSMDFF-Net achieves new state-of-the-art results. The code will be available at https://github.com/wycloveinfall/MSMDFF-NET.**

***Index Terms*— Multiscale residual module, remote sensing (RS) image, road extraction, strip convolution with direction.**

I. Introduction

As a primary component of human transportation, roads play a vital role in various fields, including urban and rural development [1], vehicle navigation [2], and crisis response [3]. With the development of society, there is a pressing need for timely and accurate road information to update geographical maps [4]. Traditional electronic maps and urban road traffic networks are mainly based on vectorized manual article maps, which are difficult to effectively maintain the accuracy and real-time performance of traffic maps [5]. To solve these problems, high-resolution remote sensing image (HRSI) has been utilized for road information extraction [6], [7] due to their cost-effectiveness and ability to cover large spatial areas comprehensively. However, the process of road data updating still requires manual intervention, and the time and labor costs involved have not significantly decreased [8]. In recent years, researchers have begun exploring ways to minimize manual costs during the road extraction process. With the development of artificial intelligence technology, using convolutional neural networks designed on high-performance GPU devices to replace manual features, has been proven to increase semantic segmentation efficiency by tens or even hundreds of times [9], [10], [11], [12], [13].

So far, numerous deep learning (DP)-based algorithms have been proposed to increase the speed of road extraction, such as Deep Road Mapper [14], Topology Loss [15], Improve Road Connectivity [16], and LinkNet [17]. Among of them, some mainstream convolutional neural networks used for road extraction are based on encoder–decoder architecture, such as the LinkNet architecture [17] proposed by Chaurasia and Culurciello. Based on the LinkNet's encoder–decoder architecture and the dilation convolution, D-LinkNet was submitted by Zhou et al. [4] as the winning algorithm in the DeepGlobe road extraction challenge [3]. These works provide valuable benchmarks for road extraction from remote sensing (RS) image and actively contribute to new developments at the intersection of computer vision and RS.

Since DP-based road extraction methods often produce fragmented roads, they pose significant challenges for practical applications. To address this issue, recent efforts have concentrated on improving the model's understanding of long-range context and multiscale information. Mosinska et al. [15] utilized U-Net combining with multiple loss functions to iteratively refine the road delineation. Liu et al. [18] integrated multilevel features, including road surfaces, edges, and centerlines, to improve road prediction. Mei et al. [19] proposed the SCM to build a connectivity attention network (CoANet), which can jointly learn segmentation and pairwise dependencies and preserve better road connectivity. Wulamu et al. [20] and Wang et al. [21] added the ASPP module after the feature extraction based on the encoder–decoder architecture to further extract the multiscale context information of the image. Chen et al. [22] used patches with different sizes to

Manuscript received 12 October 2023; revised 15 January 2024 and 27 February 2024; accepted 16 March 2024. Date of publication 25 March 2024; date of current version 1 April 2024. This work was supported by the National Natural Science Foundation of China under Grant 41571333. *(Corresponding author: Ling Tong.)*

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Digital Object Identifier 10.1109/TGRS.2024.3379988

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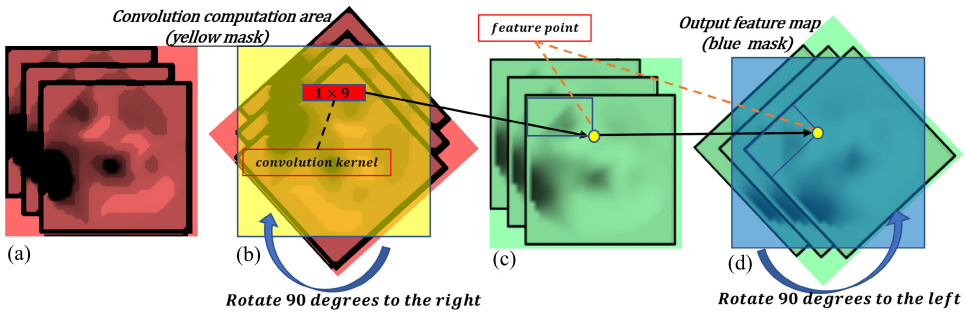


Fig. 1. Strip convolution implementation at 45°. First, the feature map [depicted by the red region in Fig. 1(a)] is rotated by 45° to obtain the result shown in Fig. 1(b). Then, a horizontal strip convolution kernel of size 1×9 is applied to convolution area [the yellow region of Fig. 1(b), which is obtained by clipping and zero-padding the red area in Fig. 1(b)]. The convolution operation produces the feature map shown in Fig. 1(c). Finally, the resulting feature map is rotated back by 45°, and cropped, zero-padded to obtain the final result feature map [the blue region in Fig. 1(d)].

train different convolutional neural network (CNN) classifiers (ResNet-50 classifiers) and merged results extracted from ResNet-50 classifiers with different scanning sizes to retrain a high-order ResNet-50 classifier for extracting the high-order context feature. Lu et al. [23] introduced a global-aware road detection network (GAMS-Net) based on multiscale residual learning. The global-aware operation was utilized to capture a wide range of feature relationships. In [24], Luo et al. emphasized the necessity of an appropriate receptive field for road extraction, introducing a high-resolution feature module (HRFM) to adapt it for roads of various scales. To some extent, these works provide valuable insights for improving the completeness of road extraction, but road fragmentation remains a serious issue. In addition, the DP-based road extraction algorithms are affected by the resolution of RS image in the experiment. In practice, the generalization ability of the DP-based model is poor. When the resolution of the input image changes, more fragmentation will be generated. Even so, this problem has not been mentioned in relevant literature.

In this article, to improve the generalization ability of the model on different resolution RS image and reduce the fragmentation of road extraction results, we propose a novel road extraction framework for RS image called multiscale and multidirection feature fusion network (MSMDFF-Net), which consists of multidirectional feature fusion initial block (MDFF-initblock), multiscale residual encoder (MSR-encoder), and multidirectional combined fusion decoder (MDCF-decoder). As mentioned earlier, square kernel convolution is limited in learning linear characteristics of the road, so we develop the strip convolution module with a direction parameter (SCM-D) as illustrated in Fig. 1. Then, we build MDFF-initblock and MDCF-decoder by using SCM-D to improve the ability of capturing long-range context information and avoid the interference of local irrelevant features. It is very useful to the integrity of road when the road is blocked by other objects. Finally, to improve the generalization of the model on different resolution RS image, we propose a novel encoder called MSR, which is designed to fuse multiscale information of input with different resolution for extracting more complete road. Our contributions are summarized as follows.

1) We propose the MSMDFF-Net for reducing the fragmentation of road extraction results, which achieves significant improvements over other methods on three different resolution datasets.
2) Considering the limitations of SCM, we develop a new method called SCM-D for calculating strip convolution in arbitrary directions, which has broader applicability. By using SCM-D, we propose MDFF-initblock and MDCF-decoder. They use the method of integrating linear features extracted by strip convolution with local features extracted by square convolution. It increases the linear receptive field of the model, making the long-distance features and local features blend with each other, which is very useful to improve the integrity of the road.
3) To enhance the model's generalization capabilities and the completeness of the extracted results, we introduce a novel encoder structure termed as the MSR-encoder. This MSR-encoder accepts data of varying resolution at distinct stages of the encoding process, thereby enabling optimal utilization of multiscale information at each encoding stage.
4) We build a pytorch-based road extraction platform. We transplant the road extraction algorithms built on other platforms to pytorch. Thus, different road extraction models can share the same experimental parameters, which makes the comparison between them more scientific and reliable.

II. Related Architecture

A. Initblock Architectures

The initialization block (Init-Block) is typically the first module of the road extraction network. It carries out preliminary feature extraction and downsampling of high-resolution RS images. In LinkNet [17] series, the encoder starts with an initial block that performs a convolution on the input image with a 7×7 kernel (stride $= 2$), followed by a 3×3 max pooling with a stride of 2 [17], [25], [26]. However, the convolution and pooling operations lead to two consecutive downsampling steps, resulting in a significant loss of information in the feature maps passed to the subsequent encoder [27]. Motivated by Inception-v4 [28], the literature of [29] defined stem-block as a stack of three 3×3 convolution layers followed by a 2×2 max pooling layer. The first conv-layer

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works with stride = 2 and the other two are with stride = 1. Inspired by deeply supervised object detectors (DSODs) [29], [30] replaced the input layers [26] with three 3×3 convolution layers (stride = 2) and one 2×2 mean pooling layer (stride = 1). These changes can evidently improve the detection performance in their experiments by reducing the information loss from raw input images compared with the original design.

The efficacy of traditional square convolutions is restricted in numerous aspects, such as the inability to account for important spatial hierarchies between features, and the lack of rotational invariance [31]. To address these limitations, Sabour et al. [32] and Hinton et al. [33] proposed the CapsuleNet. This model employs convolutional layers for feature extraction and replaces the subsequent nonlinear layers with dynamic routing and reconstruction regularization. This innovative approach allows the network to consider the spatial context of the target, ensuring it remains robust to postures or changes that may occur in the test data. However, road targets differ from other semantic segmentation targets, being spread throughout the entire RS image. As such, the square convolution kernels of Capsule Net still exhibit a limited capacity to integrate the context of these targets.

In this article, we propose the MDFF-initblock aiming to reduce the loss of road context feature at the initial stage. In MDFF-initblock, we use four SCM-D to fuse the multidirectional road context relationship, and then input it into the encoder to enhance the network's ability to extract the occlusion area of road.

B. Encoder Module

In the process of road extraction from RS imagery, the encoder is utilized to extract high-dimensional features relevant to roads. Currently, numerous classical CNNs architectures as encoder have been widely employed in various semantic segmentation tasks. These networks include visual geometry group (VGG) [34], residual network (ResNet) [25], GoogleNet [35], DeepLab [13], [36], DenseNet [26], and ResNetXt [37]. Among them, ResNet and DenseNet, are widely used in road extraction tasks. For example, LinkNet [17] and D-LinkNet [4] utilized ResNet18 as their encoder. Both of them are relatively lightweight networks that have demonstrated high precision on several benchmarks [38]. DenseNet was employed as the encoder in DensUNet [39] and CADUNet [40], which leverages dense blocks to efficiently compress models through feature reuse and enhances the performance of UNet in semantic segmentation. To accelerate model inference speed, Zhao et al. [41] proposed an image cascade network (ICNet). It inputs low-resolution images into a deeper CNN branch and high-resolution images into a lightweight CNN branch, which improves the inference speed of the model but sacrifices its accuracy. Differing from ICNet, Xie et al. [42] proposed a multiscale DenseNet (MS-DenseNet) framework, in which images of different scales are inputted in parallel into CNN branches of the same depth. It is efficient for small-scale images in terms of computational time, but for larger scale images, the computational time and graphics card memory usage increase proportionally, limiting the construction of deeper CNN networks.

In fact, these universal models for semantic segmentation often neglect the unique characteristics of roads in RS images. This oversight underscores the considerable potential that DP-based road extraction methods still possess. Regarding the improvement of road connectivity, [16] has designed a stacked multibranch module that effectively utilizes dual supervision, enabling information flow between tasks and boosting connectivity. For extracting the multiscale features and expanding the receptive field range, GAMSNet [23] incorporated MS-ResBlock (multiscale residual learning) and GA (the globally aware network) in each residual block to establish the encoder. To improve the accuracy of road extraction from RS images, SGCN [43] constructed the encoder using split depth-wise (DW) separable convolution, which enabled the extraction of channel and spatial features and improved the expression ability of road features. Although these modules improve the quality of road extraction results, they do not focus on improving the generalization ability of the model. For example, bad fragmentations are often produced when using data with different resolution. At present, it remains a thorny issue.

In practice, we find that the inference results will vary with the change of resolution when the trained model is used to extract the road in the same area, and this difference can complement each other to make the road more complete. Inspired by this, we design MSR-encoder by adding a multiscale input interface to the ResNet, which can improve the generalization of the model on different resolution RS image.

C. Decoder Module

The decoder utilizes upsampling and feature fusion operations to help capture road context and semantic information, generating pixel-level road masks or boundary images. The most sample decoder was initially introduced in fully convolutional networks (FCNs) [44]. Then, some researchers have constructed more complex decoder structures [4], [17], but these decoders are not designed according to the road's characteristics. They using a simple feature fusion approach, which is an important reason for the fragmentation of road extraction results.

Recent studies have proved that the quality of road extraction results can be improved by designing special decoders. To improve the spatial accuracy in road extraction results, BT-roadnet [45] developed a different decoder that can produce five lateral feature maps and one output feature map instead of the single-output structure. However, the improvement of the quality of road extraction is limited because it does not take into account the linear characteristics of the road.

Inspired by the shape of roads, CoANet [19] used strip convolution to design a decoder that conforms the linear characteristics of the road. The encoder was made according to the road's morphological characteristics, but does not account for the relationship between local features at the same time, leaving much room for improvement in the decoder.

Different from the decoder of CoANet, we use self-developed SCM-D and traditional square convolution to

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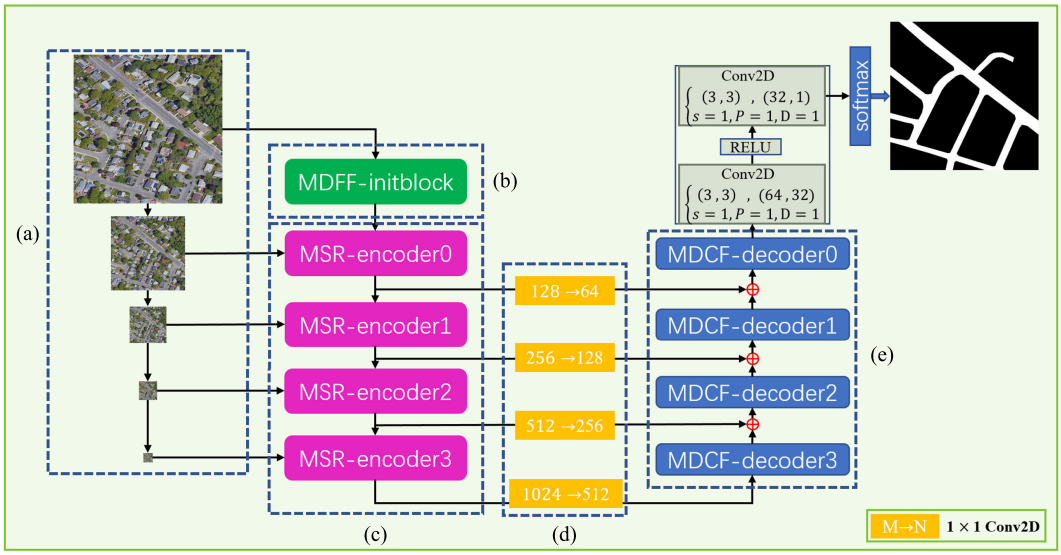


Fig. 2. Architecture of the proposed MSMDFF-Net. Part (a) involves downsampling to generate four images with different resolution. The original image is input to the MDFF-Initblock of part (b), and the other four images are input to the four encoding stages of the MSR-encoder of part (c). In part (d), a simple channel conversion is carried out to connect the MSR-encoder with the MDCF-decoder of part (e).

process high-dimensional road feature maps in parallel, and combine local features with multidirectional remote context features to further strengthen the application range of the decoder.

III. Method

In this section, we first introduce the overall structure of the proposed MSMDFF-Net. After this, three important modules in MSMDFF-Net are presented, namely, MDFF-initblock, MSR-encoder, and MDCF-decoder.

A. Multiscale and Multidirectional Feature Fusion Network

The overall architecture of the proposed MSMDFF-Net is shown in Fig. 2. As a variant of LinkNet, our MSMDFF-Net inherits the excellent structure of LinkNet, in which skip connection layers connect the encoder and the decoder. In particular, MSMDFF-Net contains the special encoder having two input branches. The entire architecture can be divided into five parts.

The MDFF-initblock (Fig. 2(b) green box) is located at the front of the encoder aiming to minimize feature loss in the early stages. The MSR-encoder (Fig. 2(c) red box) has four stages, each stage have two input branches. The left branch is connected to the four images of varying resolution from Fig. 2(a), while the upper branch receives the feature map from the previous convolutional layer. This arrangement is designed to facilitate the learning of road multiscale information by merging feature maps of different scales so that it can improve the generalization of the model on different resolution RS image. Here, the downsampling operation shown in Fig. 2(a) is performed with "torch.nn.functional.interpolate" in PyTorch, which resamples using pixel area relation. The size of the four images with different resolution are corresponding to the input of the four stages of the encoder.

In the MDCF-decoder (Fig. 2(e) blue box), the strip convolution modules in different directions are paired together to strengthen the connection between local features and remote context features, so it can directly improve the integrity of road extraction results. The yellow module [Fig. 2 (d)] represents the 1 × 1 convolution layer, which aims to convert dimension M to N without changing the size of the feature map.

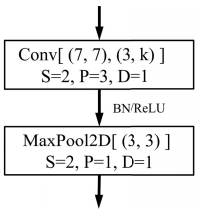


Fig. 3. Original Initblcok.

B. MDFF-Initblock

In the LinkNet series [17], [19], [23], [26], [27], InitBlock is usually used to bridge the input image and the encoder as shown in Fig. 3. Due to Init-Block setting the convolution kernel in a square shape and having two consecutive down channels, the ability to learn the long-distance linear features of the road is limited. Additionally, the InitBlock tends to lose contextual features in road occluded regions, leading to fragmentation in the road extraction results.

Fortunately, strip convolution is consistent with the characteristics of road shape which can reduce the interference of irrelevant information [46]. Inspired by this, we develop the SCM-D, as shown in Fig. 1, to solve the defect that traditional strip convolution cannot change the direction of calculation. SCM-D is used to build the MDFF-initblock to enhance the learning of remote context features with less feature loss, as illustrated in Fig. 4.

First, MDFF-initblock passes through square convolution layer [kernel size is 3 × 3, output is $O_1(f)$] and four strip convolution layers (kernel size is 1 × 9), respectively. Second,

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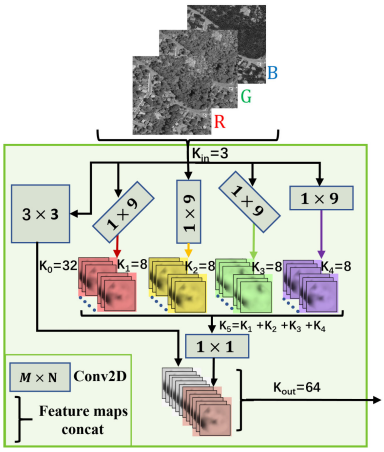


Fig. 4. Multidirectional features fusion initblock (MDFF-initblock).

the output features of four directions are fused by using the convolution layer [kernel size is 1×1, output is $O_2(f)$]. Finally, the fused feature maps $[O_2(f)]$ of the four directions are cascaded with the feature maps $[O_1(f)]$ from the square convolution layer, as the output $[O_s(f)]$ of the convolution module. The above operation can be described as shown in the following formulas:

$$O_1(f) = G_{\blacksquare}^{3*3}(f) \tag{1}$$

$$O_2(f) = G_{\blacksquare}^{1*1}\left(F_c[G_{\nearrow}^{1*9}(f), G_{\uparrow}^{1*9}(f), G_{\searrow}^{1*9}(f), G_{\rightarrow}^{1*9}(f)]\right) \tag{2}$$

$$O_s(f) = F_c[O_1(f), O_2(f)] \tag{3}$$

where $G_{\blacksquare}^{M*N}$ and $G_{\nearrow\uparrow\searrow\rightarrow}^{M*N}$ denote a standard square convolution and strip convolution with direction($\nearrow\uparrow\searrow\rightarrow$), respectively. F_c represents channel wise concatenation. Unless otherwise specified, every convolutional layer mentioned in this article is followed by a batch normalization (BN) layer and a ReLU activation layer.

C. MSR-Encoder

The existing DP-based road extraction methods hardly consider the impact of the resolution of RS image on the performance of the algorithm. However, the spatial characteristics of roads are not the same at different resolution. As presented in Fig. 5, when the trained model is applied on the same RS image, the completeness and accuracy of the extraction results at different resolution are different. The missing roads may be well extracted at another resolution (indicated by the red regions). It also can be seen that new additions to missing parts of roads (indicated by the green regions) at another resolution. By integrating the extraction results from different resolution, the completeness of road extraction can be greatly improved as Fig. 5(e).

Based on the above results, MSR-encoder is proposed to improve the integrity of road extraction and the generalization of the model to data of different resolution. The MSR-encoder can receive RS images of multiple resolution and enable the network to internally fuse multiscale features. The coding process of the whole network includes four coding modules, one of which is illustrated in Fig. 6.

TABLE I

CONFIGURATION OF DIFFERENT ENCODING STAGES. EACH STAGE OF MSR-ENCODER CONTAINS TWO BRANCHES, RESNET34.LAYER AND RESBLOCK. THE FINAL OUTPUT SIZE IS OBTAINED BY CONCATENATING THE OUTPUT SIZES OF THE TWO BRANCHES

Encoder_i	Submodule	Parameters	Submodule Outputsize	Final Outputsize
MSR-encoder0	ResNet34.layer1	$\begin{bmatrix}3\times3,64\\3\times3,64\end{bmatrix}\times3$	256×256×64	64×64 ×128
	ResBlock(K1)	$K_1 = 64$	256×256×64	
MSR-encoder1	ResNet34.layer2	$\begin{bmatrix}3\times3,64\\3\times3,64\end{bmatrix}\times4$	128×128×128	128×128 ×256
	ResBlock(K1)	$K_1 = 128$	128×128×128	
MSR-encoder2	ResNet34.layer3	$\begin{bmatrix}3\times3,64\\3\times3,64\end{bmatrix}\times6$	64×64×256	64×64 ×512
	ResBlock(K1)	$K_1 = 256$	64×64×256	
MSR-encoder3	ResNet34.layer4	$\begin{bmatrix}3\times3,64\\3\times3,64\end{bmatrix}\times3$	32×32×512	32×32 ×1024
	ResBlock(K1)	$K_1 = 512$	32×32×512	

In Fig. 6, the ith encoder consists of two parts. 1) ResNet34.layer(i) [25] is used to receive the feature map of the previous module, which can reuse feature layer to train deeper network. 2) The ResBlock is applied to get the scale information of the road at different stages of the encoder. The input to the ResBlock is the downsampled version of the original RS image. It is important to note that at different encoding stage, the ResBlock of the encoder accepts images of different sizes, as illustrated in Fig. 2. Then, the feature maps of the above two parts are cascaded out. In this way, the original input image is downsampled and then is sent to the encoder. Hence, the different scale information of the road can be used by each stage of the encoder. The detailed input and output feature map sizes for each encoder in the network are listed in Table I.

D. MDCF-Decoder

The 1-D filters are more aligned with road shapes than square filters. Replacing the 3×3 transpose convolution in the decoder by 1-D transpose convolution gives better accuracy [47]. Otherwise, the 1-D filters also has been used to construct the decoder in [19] to capture long-range contextual relationships by directly fusing the context features of the four directions. However, roads do not only have long linear features, and the local range features of road intersections are also important. Simply using the contextual relationships of roads may lead to road fractures at intersections.

In order to reduce the fragmentation in road intersections and road areas covered by other objects, both SCM-D and square convolution are combined to construct MDCF-decoder, which fuses local features with remote context features. The whole decoding process of the network contains four decoding modules, which are connected to the four encoders in Fig. 2. The ith decoding module is shown in Fig. 7.

The MDCF-decoder adapt the SCM-D with four directions (horizontal, vertical, left diagonal, and right diagonal) to learn

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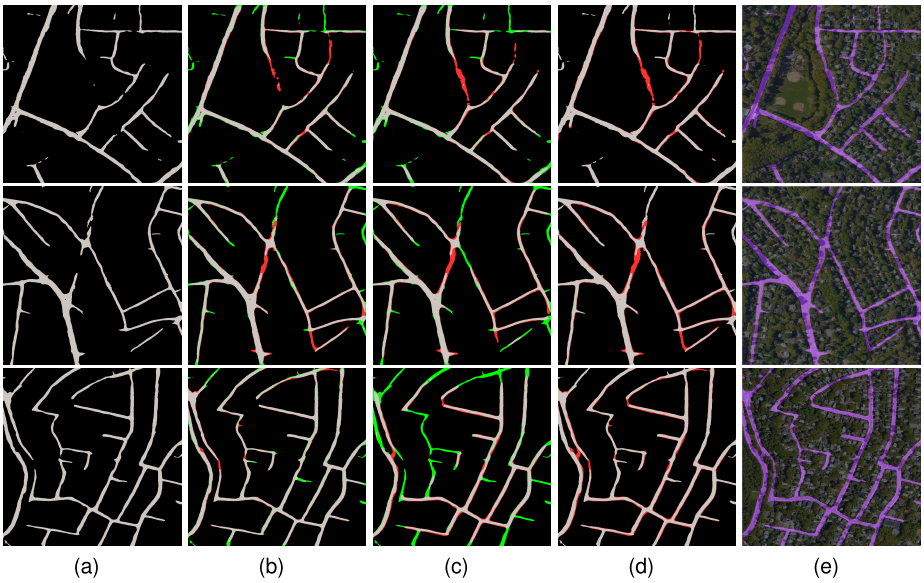


Fig. 5. Prediction results obtained by the same model at different resolution. We selected three different regions from LSRV dataset [23] and reduce their resolution according to the scale factor of 0.95, 0.8, and 0.75. Then, these images were input into the LinkNet50 trained with DeepGlobe dataset [3], and the inference results at four different resolution as shown in Fig. 5(a)–(d), respectively. Finally, after resolution adjustment, the other three results were respectively performed "pixel or" operation with the output of the original resolution, as shown in Fig. 5(e). Using the results in Fig. 5(a) as contrast, green represents reduced roads, red represents increased roads, and purple represents the result of fusion.

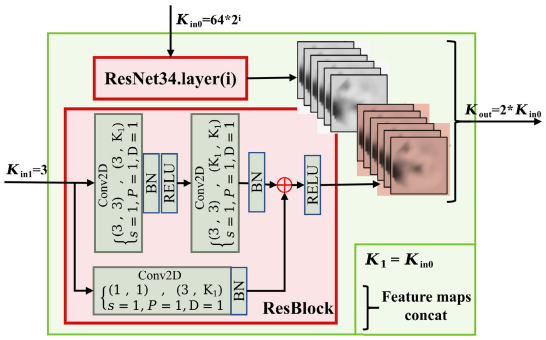


Fig. 6. MSR-encoder. The encoder module has two inputs: the position of K_{in0} and K_{in1}. Here, Ki indicates the number of channels of the feature map, and i represents the ith encoder module, $i = 0, 1, 2, 3$.

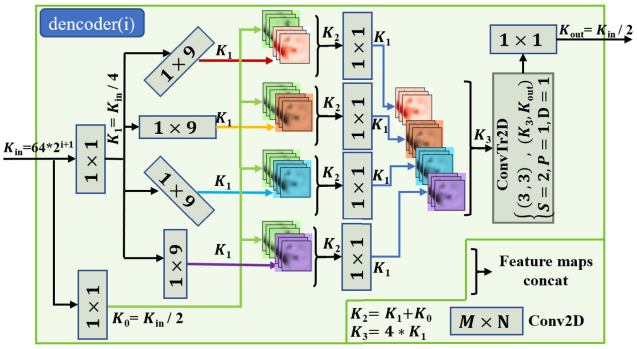


Fig. 7. Structure of MDCF. Where i represents the ith decoding module, $i = 0, 1, 2, 3$. Ki represents the number of channels in the feature map. All the 1-D convolution operation $S = 1$, $D = 1$, $P = 0$.

linear features of the road at longer distances. The features of the strip convolution are paired and cascaded with the features of the square convolution separately, to capture the more complex topological relationships of the roads. After passing through a 1×1 convolutional layer, the output results are $F^{MF}_{\nearrow\uparrow\searrow\rightarrow}(f)$. Then, the output $O^t(f)$ of the deconvolution fuses the long-range contextual relations to reduce the influence of local interference features. Finally, the output $O^t(f)$ passes through a 1×1 convolution kernel for channel fusion. The output $O^c(f)$ can be calculated by the following formula:

$$F^{MF}_{\nearrow}(f) = G^{1*1}_{\blacksquare}\left(F_c[G^{1*1}_{\blacksquare}(f), G^{1*9}_{\nearrow}\left(G^{1*1}_{\blacksquare}(f)\right)]\right) \tag{4}$$

$$F^{MF}_{\uparrow}(f) = G^{1*1}_{\blacksquare}\left(F_c[G^{1*1}_{\blacksquare}(f), G^{1*9}_{\uparrow}\left(G^{1*1}_{\blacksquare}(f)\right)]\right) \tag{5}$$

$$F^{MF}_{\searrow}(f) = G^{1*1}_{\blacksquare}\left(F_c[G^{1*1}_{\blacksquare}(f), G^{1*9}_{\searrow}\left(G^{1*1}_{\blacksquare}(f)\right)]\right) \tag{6}$$

$$F^{MF}_{\rightarrow}(f) = G^{1*1}_{\blacksquare}\left(F_c[G^{1*1}_{\blacksquare}(f), G^{1*9}_{\rightarrow}\left(G^{1*1}_{\blacksquare}(f)\right)]\right) \tag{7}$$

$$O^t(f) = G^{3*3}_{\mathbf{d}}\left(F_c\left[F^{MF}_{\nearrow}(f), F^{MF}_{\uparrow}(f), F^{MF}_{\searrow}(f), F^{MF}_{\rightarrow}(f)\right]\right) \tag{8}$$

$$O^c(f) = G^{1*1}_{\blacksquare}\left(O^t(f)\right) \tag{9}$$

where G^{M*N}_{d} a deconvolutional kernel with a size of M*N.

IV. Experiments

In this section, we provide detailed instructions on three datasets and four evaluation metrics employed in the experiments. Additionally, detailed evaluation results in both quantitative and qualitative are presented.

In order to ensure that the comparison of evaluation index parameters between different methods is more scientific and reliable, and to enhance the repeatability of the article, we build a pytorch-based road extraction platform. We transplant road extraction algorithms built on other platforms to pytorch so that all experiments can use same environmental parameters. The specifical parameter settings and source code of all the experiments will be described in detail.

A. Datasets

We evaluate our approach against several other state-of-the-art methods on three datasets with different resolution (Massachusetts [48], DeepGlobe [3], SpaceNet [49]).

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1) Massachusetts Road-dataset [1 m/pixel]: It contains 1171 aerial images, with a size of 1500 × 1500 pixels per image. The target map is generated from the centerlines of roads obtained by rasterizing the OpenStreetMap project, with a linewidth of seven pixels and all images rescaled to a resolution of 1 m/pixel. We randomly selected 995 (85%) images as the training set and 176 (15%) images as the test set. Each 1500 × 1500 image is converted into a 1536 × 1536 image by padding with 18 rows and 18 columns of zero-value pixels on the top, bottom, left, and right sides. For the training set, each image is divided into 25 pixel-blocks of size 512 × 512, with a 256-pixel overlap in both the horizontal and vertical directions. For the test set, each image is segmented into nine pixel-blocks of size 512 × 512, with no overlap between adjacent blocks.

2) DeepGlobe dataset [0.5 m/pixel]: The dataset consists of 6226 training images, 1243 validation images, and 1101 test images, with each image having a size of 1024 × 1024 pixels. We divide the 6226 training images into 5292 (85%) for training process and 934 (15%) for testing process. Each 1024 × 1024 image in the training process is split into nine pixel-blocks of size 512 × 512, with a 256 pixel overlap in both the horizontal and vertical directions. Each image is divided into four pixel-blocks of size 512 × 512 in the testing process, with no pixel overlap between the blocks.

3) SpaceNet dataset [0.3 m/pixel]: The dataset consists of 2780 images with size 1300 × 1300, 2213 (80%) are used for training stage and 567 (20%) for testing stage. The ground truth is given in the form of a line-string that indicates the road's centerline. To generate road labels with a width of 7 m, we follow the same approach as the Massachusetts Road dataset, with a linewidth of 24 pixels at the given resolution. The 1300 × 1300 image is resized to 1200 × 1200 by reducing 50 rows and 50 columns from the top, bottom, left, and right. In the training set, the image is divided into 16 patches of size 512 × 512 pixels, with an overlap of 256 pixels in both the horizontal and vertical directions. In the test set, the image is divided into four patches of size 650 × 650 pixels, with no overlap between the patches.

B. Evaluation Metrics

1) Pixel-Based Metrics: To evaluate the performance of our method for road segmentation, we use precession (P), recall (R), F1-score (F1), and intersection over union (IoU) metrics

$$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}} \tag{10}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}} \tag{11}$$

$$F1 = \frac{2\text{TP}}{2\text{TP}+\text{FP}+\text{FN}} \tag{12}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP}+\text{FP}+\text{FN}} \tag{13}$$

where P, N, TP, FP, and FN represent the positive, negative, true positive, false positive, and false negative pixels in the prediction map, respectively.

TABLE II
ENVIRONMENTAL PARAMETERS. ALL OF THE ROAD EXTRACTION NETWORKS MENTIONED IN THIS ARTICLE USE THE SAME ENVIRONMENTAL PARAMETERS BELOW UNLESS OTHERWISE SPECIFIED

System	Ubuntu22.04.3 LTS
GPU Resource	NVIDIA Tesla P40 * 3
Driver Version	515.105.01
CUDA Version	11.8
DL Framework	Pytorch 1.8.1
Compiler	Python 3.8.16
Optimizer	Adam
Learning Rate (LR)	0.005
LR Policy	PolyLR
Loss Function	Ldice and Lbce
Train and Test (seed)	85% 15% (8)
Batch Size	6

2) Graph-Based Metric: The average path length similarity (APLS) [49] is used in our experiments to evaluate topological correctness and connectivity of roads

$$C = 1 - \frac{1}{N}\sum \min\left\{1, \frac{|L(a,b) - L(a',b')|}{L(a,b)}\right\} \tag{14}$$

where N = number of unique paths, $L(a, b)$ = length of path(a, b). The sum is taken over all possible source (a) and target (b) nodes in the ground truth graph. The node a' and b' denote the node in the proposal graph closest to the location of ground truth node a and b, respectively.

C. Implementation Details

All the road extraction networks experimented in this article were based on the DP framework PyTorch [50], which were carried out on 3 × NVIDIA Tesla P40 with 24GB memory. In the process of training, adaptive moment estimation (Adam) was used as the optimizer. The three datasets were divided into training and test set according to the proportion of 85% and 15%, using sklearn.model_selection.train_test_split with the random seed 8. The data augmentation includes random rotation (30°,60°,120°,150°), vertical and horizontal flipping. We used the "poly" strategy to update the learning rate following [19]. The sum of the binary cross-entropy (BCE) loss and dice coefficient loss was adopted as the loss function, and we used 0.5 as the threshold to generate the final output. These experimental parameters were organized as shown in Table II

$$L_{\mathbf{BCE}} = -\frac{1}{N}\Sigma_{i=1}^{N}[y_i \cdot \log(\hat{y}_i) + (1-y_i)\cdot\log(1-\hat{y}_i)] \tag{15}$$

$$L_{\mathbf{Dice}} = \frac{2\sum_{i=1}^{N}(y_i\hat{y}_i)}{\sum_{i=1}^{N} y_i^2 + \sum_{i=1}^{N}\hat{y}_i^2} \tag{16}$$

$$\text{Loss} = (1 - L_{\mathbf{Dice}}) + L_{\mathbf{BCE}} \tag{17}$$

where N indicates the number of elements in a H × W slice, y_i is the ground truth denoting road or background for a

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given pixel in position i, and $\hat{y}_i$ is the corresponding predicted probability of the segmentation branch.

To ensure the fairness and scientific rigor of the comparative experiments, all experiments involving the road extraction algorithms mentioned in this article are conducted using the same experimental parameters as mentioned above unless otherwise specified.

D. Open-Source Model Code

To ensure convenient and impartial comparison of the performance differences among different models, we dedicate a significant amount of time to debugging all the methods on the PyTorch platform. All road extraction networks share a common training and testing framework.

For early networks originally implemented in Caffe or TensorFlow, we performed cross-platform porting, such as UNet++, DeepRoadMapper, and RoadCNN. For recent networks without available source code, we reproduced the code based on the corresponding literature, such as GAMSNet and CADUNet. For approaches that provide pytorch code directly, we used only the code for the network architecture, such as SGCNNet, CoANet, LinkNet series, and D-LinkNet series. The code for all the networks used in the experiment will be organized into a complete demo and made available at the following link https://github.com/wycloveinfall/MSMDFF-NET.

E. Comparison With State-of-the-Art Methods

In order to validate the effectiveness of the proposed model on RS images with different resolution, three datasets with different resolution are used in this section: the Massachusts dataset (1 m/pixel), the Deepglobe subset dataset (0.5 m/pixel), and the SpaceNet dataset (SP) (0.3 m/pixel).

Quantitative and qualitative comparative experiments are conducted with several classical mainstream network architectures, including DeepRoadMapper [14], RoadCNN [8], UNet++ [12], LinkNet34, LinkNet50 [17], DlinkNet34, DlinkNet50 [4], and the latest state-of-the-art road extraction methods GAMSNet [23] and CoANet [19]. DeepRoadMapper and CoANet use additional network structures to improve the completeness of road extraction results. In order to fairly compare the performance of the network structures, we only use the pixel segmentation branch.

1) Quantitative Analysis: The quantitative experimental results of our proposed MSMDFF-Net compared to several other state-of-the-art networks on the Massachusetts dataset (MA), Deepglobe dataset (DP), and SP are shown in Tables III–V, respectively. In order to analyze the ability of the model proposed in this article to solve the fragmentation problem of road extraction results, we use the APLS metric as the evaluation index of road connectivity. The APLS of road extraction results are shown in Table VI. It is easy to see that the evaluation metrics given by our MSMDFF-Net are at the state-of-the-art level compared to other networks. Regarding the P of Table III, the MSMDFF-Net does not achieve the highest value, which is the same as the result of P evaluation in Table V. That is because MA and SP are

TABLE III

QUANTITATIVE COMPARISON OF OUR PROPOSED MSMDFF-NET WITH SEVERAL OTHER STATE-OF-THE-ART ROAD EXTRACTION METHODS ON THE **MA**. THE BOLD VALUES IN EACH COLUMN REPRESENT THE HIGHEST VALUES, AND RED INDEX DENOTING THE RANKING OF THE METRIC

Index	Methods	P (%)	R (%)	F1 (%)	Road IOU (%)
A	UNet++	**81.79**[1]	60.69[11]	69.68[6]	53.47[6]
B	DeepRoadMapper	77.62[2]	64.59[5]	70.51[2]	54.45[2]
C	LinkNet34	75.95[5]	64.35[6]	69.67[7]	53.46[7]
D	LinkNet50	74.71[8]	63.43[10]	68.61[10]	52.22[10]
E	DlinkNet34	76.23[4]	64.26[7]	69.73[4]	53.53[4]
F	DlinkNet50	75.10[6]	65.59[3]	70.02[3]	53.87[3]
G	RoadCNN	77.00[3]	63.67[8]	69.71[5]	53.50[5]
H	GAMSNet	73.44[10]	64.90[4]	68.90[9]	52.56[9]
I	CoANet	72.78[11]	66.39[2]	69.44[8]	53.18[8]
J	CADUNet	59.84[12]	50.45[12]	54.75[12]	37.69[12]
K	SGCNet	74.05[9]	63.76[9]	68.52[11]	52.12[11]
L	MSMDFF-Net	74.80[7]	**69.14**[1]	**71.86**[1]	**56.07**[1]

TABLE IV

QUANTITATIVE COMPARISON OF OUR PROPOSED MSMDFF-NET WITH SEVERAL OTHER STATE-OF-THE-ART ROAD EXTRACTION METHODS ON THE **DEEPGLOBE SUBSET DATASET**. THE BOLD VALUES IN EACH COLUMN REPRESENT THE HIGHEST VALUES, AND RED INDEX DENOTING THE RANKING OF THE METRIC

Index	Methods	P (%)	R (%)	F1 (%)	Road IOU (%)
A	UNet++	76.42[5]	74.44[7]	75.41[5]	60.53[5]
B	DeepRoadMapper	74.96[9]	75.40[4]	75.18[7]	60.23[7]
C	LinkNet34	77.86[2]	74.36[8]	76.07[3]	61.38[3]
D	LinkNet50	76.21[6]	68.30[11]	72.04[10]	56.30[10]
E	DlinkNet34	75.79[8]	73.48[9]	74.62[9]	59.52[12]
F	DlinkNet50	77.09[3]	74.73[5]	75.89[4]	61.15[4]
G	RoadCNN	76.15[7]	74.50[6]	75.31[6]	60.40[6]
H	GAMSNet	74.64[10]	75.55[3]	75.09[8]	60.11[8]
I	CoANet	76.560[4]	76.558[2]	76.559[2]	62.020[2]
J	CADUNet	69.54[12]	72.69[10]	71.08[11]	55.14[11]
K	SGCNet	72.78[11]	66.98[12]	69.76[12]	53.57[12]
L	MSMDFF-Net	**78.92**[1]	**76.85**[1]	**77.87**[1]	**63.76**[1]

generated from the center line expanded by equal width and do not cover the complete road, so that the more extracted roads lead to the lower P-value. Moreover, the proposed method has the highest scores in the four evaluation indicators P, R, F1, and IoU, which are 1.36%, 0.38%, 1.71%, and 2.81% higher than the second-best CoANet (as shown in Table IV), respectively. These suggest that the model proposed in this article provides superior pixel segmentation compared to other models. The higher recall value indicates that our model can extract a larger proportion of the true road. The F1 score, acting as a comprehensive evaluation index, better balances between precision and recall. Furthermore, a higher IOU index represents improved pixel integrity in the road extraction results.

The topological connectivity of roads, representing the degree of fragmentation of road extraction results, is an

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TABLE V

QUANTITATIVE COMPARISON OF OUR PROPOSED MSMDFF-NET WITH SEVERAL OTHER STATE-OF-THE-ART ROAD EXTRACTION METHODS ON THE **SP**. THE BOLD VALUES IN EACH COLUMN REPRESENT THE HIGHEST VALUES, AND RED INDEX DENOTING THE RANKING OF THE METRIC

Index	Methods	P (%)	R (%)	F1 (%)	Road IOU (%)
A	UNet++	**76.30**[1]	66.91[11]	71.30[8]	55.40[8]
B	DeepRoadMapper	73.15[5]	69.87[7]	71.47[7]	55.61[7]
C	LinkNet34	72.35[9]	72.73[3]	72.54[3]	56.91[3]
D	LinkNet50	76.09[2]	63.59[12]	69.28[11]	53.00[11]
E	DlinkNet34	72.85[7]	72.55[4]	72.70[2]	57.10[2]
F	DlinkNet50	73.03[6]	70.56[6]	71.77[5]	55.97[5]
G	RoadCNN	73.94[4]	69.59[8]	71.70[6]	55.88[6]
H	GAMSNet	67.63[11]	74.36[2]	70.83[9]	54.84[9]
I	CoANet	72.68[8]	71.44[5]	72.05[4]	56.31[4]
J	CADUNet	65.07[12]	67.48[10]	66.25[12]	49.53[12]
K	SGCNet	69.96[10]	68.75[9]	69.35[10]	53.08[10]
L	MSMDFF-Net	73.96[3]	**76.03**[1]	**74.98**[1]	**59.98**[1]

TABLE VI

APLS(%) COMPARISON OF OUR PROPOSED MSMDFF-NET WITH SEVERAL OTHER STATE-OF-THE-ART ROAD EXTRACTION METHODS ON THE **MA**, **DEEPGLOBE SUBSET DATASET (DP)**, AND **SP**

Index	Methods	MA	DP	SP
A	UNet++	59.25[2]	64.50[9]	59.04[4]
B	DeepRoadMapper	57.01[4]	65.90[7]	58.50[6]
C	LinkNet34	57.16[3]	67.92[4]	61.45[2]
D	LinkNet50	55.14[9]	62.82[10]	55.63[11]
E	DlinkNet34	55.11[10]	67.96[3]	61.27[3]
F	DlinkNet50	56.13[7]	67.47[5]	58.65[5]
G	RoadCNN	55.42[8]	65.63[8]	57.84[8]
H	GAMSNet	56.67[6]	67.00[6]	58.29[7]
I	CoANet	56.80[5]	68.69[2]	57.38[9]
J	CADUNet	25.13[12]	61.46[11]	47.09[12]
K	SGCNet	49.53[11]	57.39[12]	55.73[10]
L	MSMDFF-Net	**60.27**[1]	**69.78**[1]	**62.01**[1]

important evaluation index of road extraction results. Our proposed MSMDFF-Net achieves the highest APLS evaluation index (60.27%, 69.78%, and 62.01%) on three different resolution datasets (MA, DP, and SP datasets), outperforming other algorithms significantly. This suggests that the network using SCM-based decoder modules has obvious advantages over other algorithms. We build the decoder in parallel using square convolution and SCM-D, which can fuse remote context with local range features. This fusion operation can not only use the remote contextual features of the road, but also enhance the use of local features, so the model can extract a more complete road.

In addition, comparing the algorithms of different rankings in Tables III–VI, DeepRoadMapper is the second-best algorithm on MA (1 m/pixel). CoANet obtains the second-best results on DeepGlobe dataset(0.5 m/pixel). DlinkNet ranks second on the SpaceNet dataset(0.3 m/pixel). These comparisons show that different methods have different adaptability to RS images of different resolution. The conclusion can also be drawn by comparing the third-best or fourth-best methods in Tables III–VI. As we mentioned earlier, the performance of road extraction algorithms can be affected by the resolution of RS images. On the whole, our proposed MSMDFF-Net demonstrates superior performance on the three datasets, which shows that our method has good adaptability to different resolution datasets. This achievement can be attributed to the proposed MSR-encoder. In the coding stage, by dividing the input image into several different resolution and inputting them into the encoder, the multiscale information of roads can be fully utilized to integrate the road features of different scales, which makes the multiscale features complement each other. So, it improves the compatibility of the network to the images of different resolution.

2) Qualitative Analysis: The qualitative comparison results of our MSMDFF-Net and other methods are illustrated in Fig. 8. Several different typical regions are selected, which show three examples of MA [48], three examples of DeepGlobe dataset [3] and two examples of SP [49].

In the multilane region with complex topological relationship (the (a) column in Fig. 8), the road extraction results obtained by MSMDFF-Net have the most complete connectivity compared with other methods. In the housing dense areas (the (b) and (c) column in Fig. 8), the road background is very complex, and many algorithm extraction results are broken at the intersection of roads, while our proposed method can extract most of the road intersections. When the color of the road is very similar to the background color, the color and texture of the road are very weak [(d) and (e) columns in Fig. 8], which is difficult to fully extract the road target. In addition, occlusion will also cause fragmentation of road extraction results. For example, in Fig. 8(f), when some roads are divided into discontinuous sections by trees, roads extracted by other algorithms produce different degrees of fracture, while our proposed MSMDFF-Net can solve this problem well. Moreover, in Fig. 8(h), some green patches appear in the upper part of all the road extraction results. And the results produced by other algorithms have a large number of holes and isolated noise points. In contrast, the green road region extracted by the MSMDFF-Net has a clear shape without any breaks and retains the complete road topology. From the above experimental results, it can be seen that MSMDFF-Net can extract the most complete road in various complex scenarios.

3) Summary of Quantitative Experimental Results: Comparing the four quantitative experimental results in Tables III–VI, it is noticeable that although our method achieves the best results on the three datasets, the performance on the datasets with different resolution is different. To explore these differences more deeply, we rank the metrics of our method on three datasets as reported in Table VII.

That can be attributed to the follow two main reasons.

First, the labeling format of the DeepGlobe dataset is different from the SpaceNet and MA. The labels in the DeepGlobe dataset are manually created by annotating along the actual edges of the roads, which makes DP-based road extraction

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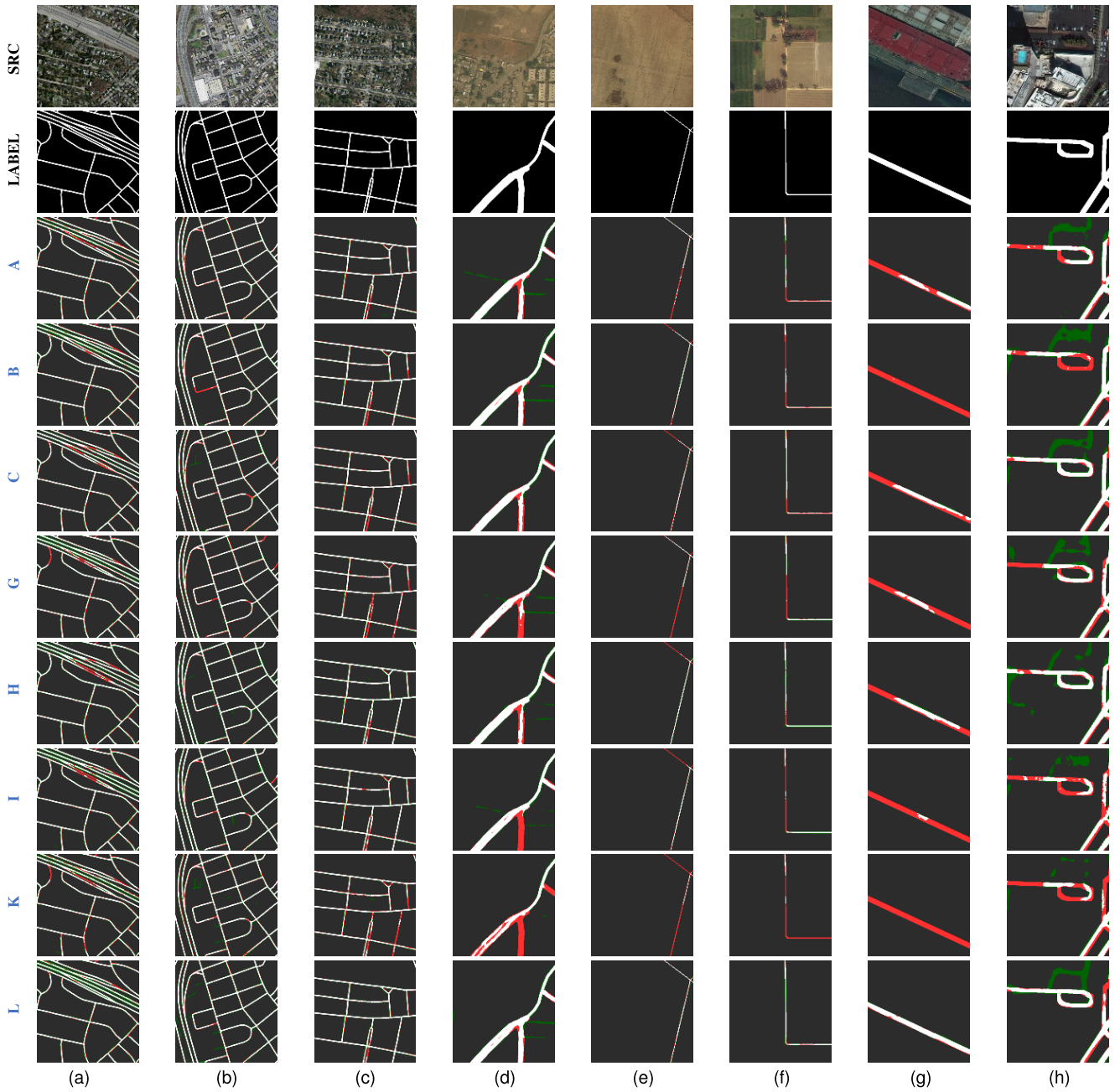


Fig. 8. Qualitative comparison of our method with several other state-of-the-art methods on the Massachusetts, DeepGlobe, and SpaceNet datasets. White color represents true positives, green color represents false positives, and red color represents false negatives. (SRC) Satellite imagery. (LABEL) Ground truth. (A)-(L) Road extraction results of UNet++ [12], DeepRoadMapper [14], LinkNet34 [17], RoadCNN [8], ImprovedConnectivity [16], GAMSNet [23], CoANet [19], SGCNet [43], and our MSMDFF-Net. (a) MA. (b) MA. (c) MA. (d) DP. (e) DP. (f) DP. (g) SP. (h) SP.

results more accurate. However, the SpaceNet and MA provide road centerline labels, and the labels required for training are generated based on the centerlines with fixed widths. The generated road masks may not cover the entire road accurately, leading to lower scores in the evaluation metrics.

Second, comparing the performance of our proposed model on the SpaceNet and MA, we observe a better result on the higher resolution SP. This is due to the fact that the input RS images are downsampled at different scales at different coding stages. This downsampling has different effects on the width and length of the road at different resolution. When the road width is less than one pixel in the feature map, the spatial information of the road is lost. Therefore, our proposed method performs better on high-resolution data under the same conditions.

F. Experiments on the Large-Scale Images

In order to verify the road extraction quality of our proposed model on large-scale RS images, we conduct experiments on large-scale areas from Massachusetts and Boston. The data are available at https://github.com/wycloveinfall/large_scale_images. We also analyzed the relationship between road extraction results and RS image resolution, testing different resolution images with different models. These models,

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Fig. 9. Visual prediction results of Massachusetts obtained by the MSMDFF-Net^{MA}.

TABLE VII

RANKED METRICS OF OUR MSMDFF-NET ON THREE DATASETS. THE BEST RESULT IS ON THE DP DATASET WITH A RESOLUTION OF 0.5 M/PIXEL, FOLLOWED BY THE SP DATASET WITH A RESOLUTION OF 0.3 M/PIXEL, AND THE MA DATASET WITH A RESOLUTION OF 1 M/PIXEL

Rank	Datasets	P (%)	R (%)	F1 (%)	Road IOU (%)	APLS (%)
1	DP	78.92[1]	76.85[1]	77.87[1]	63.76[1]	69.78[1]
2	SP	73.96[3]	76.03[2]	74.98[2]	59.98[2]	62.01[2]
3	MA	74.80[2]	69.14[3]	71.86[3]	56.07[3]	60.27[3]

MSMDFF-Net^{MA}, MSMDFF-Net^{DP}, and MSMDFF-Net^{SP}, are trained on MA (1 m/pixel), DP (0.5 m/pixel), and SP (0.3 m/pixel) datasets, respectively.

Fig. 9 shows the visual results of the proposed MSMDFF-Net^{MA} method on the Massachusetts image, which has a range of 14 116 × 16 273 pixels and a resolution of 1 m/pixel. Fig. 10 displays the visual results of MSMDFF-Net^{DP} on the Boston image, where the image has a range of 23 104 × 23 552 pixels and a resolution of 0.44 m/pixel. It is worth noting that the Boston image does not participate in any training process. The visual extraction results show that our proposed network can extract more complete roads on large-scale remote sensing images and exhibit good generalization ability on new experimental data.

Table VIII presents the quantitative experimental results of three different resolution models on two different resolution images. It shows that MSMDFF-Net^{MA} has the highest scores in the four evaluation indicators: R, F1, IoU, and APLS, for the Massachusetts (1 m/pixel) region. However, MSMDFF-Net^{MA} barely extracts any roads in Boston (0.44 m/pixel). Similarly, MSMDFF-Net^{DP} shows better road extraction performance in Boston (0.44 m/pixel) compared to Massachusetts(1 m/pixel). This indicates that the road extraction results can be influenced by the image resolution of the training set used by the model.

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TABLE VIII

QUANTITATIVE EXPERIMENTAL RESULTS OF THREE DIFFERENT RESOLUTION MODELS ON TWO DIFFERENT RESOLUTION IMAGES. MSMDFF-NET^{MA}, MSMDFF-NET^{DP}, AND MSMDFF-NET^{SP} WERE TRAINED ON MA (1 M/PIXEL), DP (0.5 M/PIXEL), AND SP (0.3 M/PIXEL) DATASETS, RESPECTIVELY

Image	Methods	P (%)	R (%)	F1 (%)	Road IOU (%)	APLS (%)
Massachusetts image	MSMDFF-Net^{MA}	79.70[1]	79.97[1]	79.83[1]	66.44[1]	76.97[1]
	MSMDFF-Net^{DP}	57.90[2]	68.43[2]	62.73[2]	45.70[2]	54.61[2]
	MSMDFF-Net^{SP}	31.40[3]	43.04[3]	36.31[3]	22.18[3]	26.85[3]
Boston image	MSMDFF-Net^{MA}	89.95[1]	08.95[3]	16.28[3]	08.86[3]	08.25[3]
	MSMDFF-Net^{DP}	85.10[2]	72.21[1]	78.12[1]	64.10[1]	61.46[1]
	MSMDFF-Net^{SP}	69.06[3]	67.95[2]	68.50[2]	52.09[2]	46.41[2]

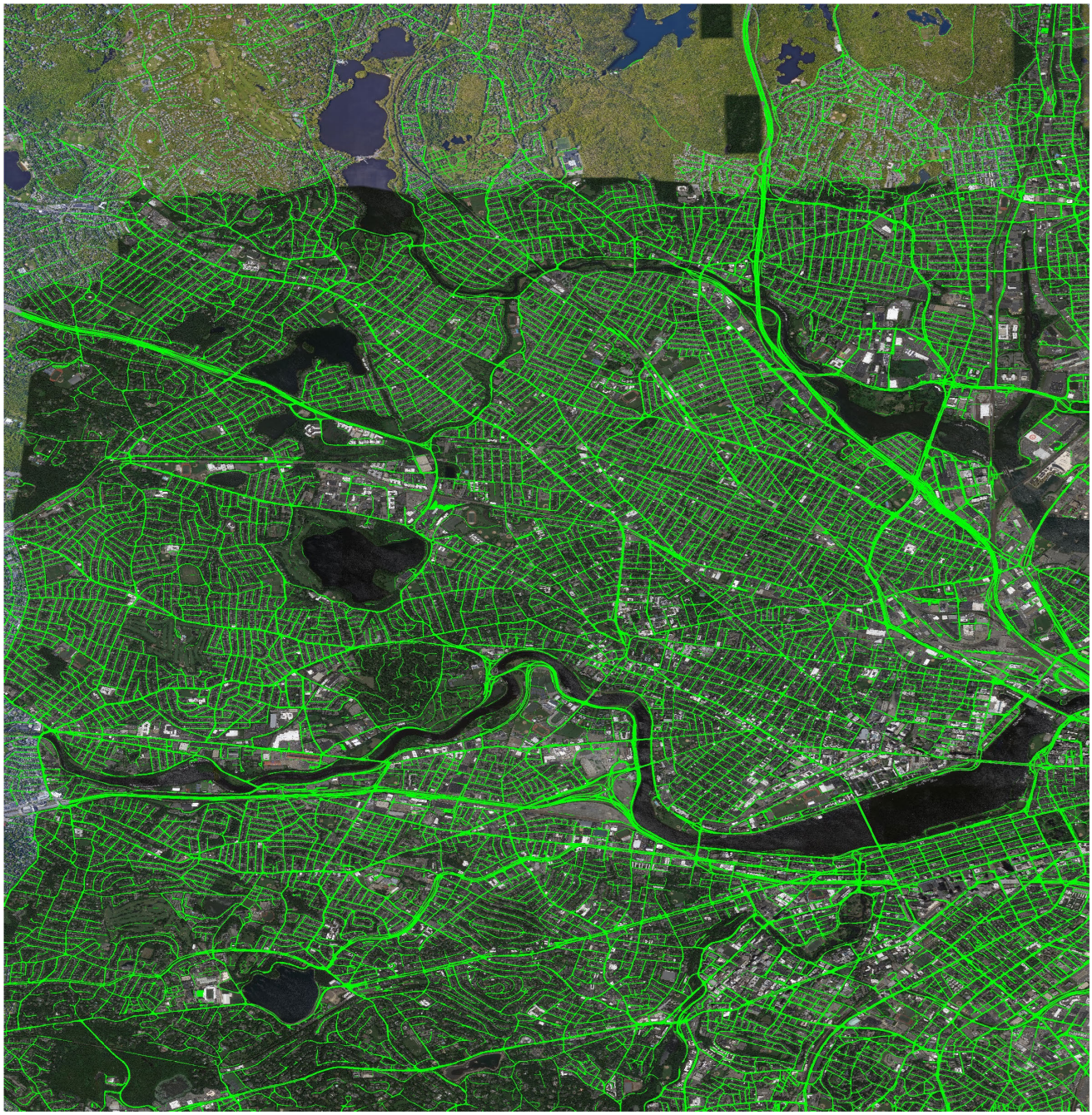


Fig. 10. Visual prediction results of Boston obtained by the MSMDFF-Net^{DP}.

For visually comparing this effect, we randomly enlarge three local areas from Figs. 9 and 10, comparing the details in Figs. 11 and 12, respectively. In Fig. 11, MSMDFF-Net^{MA} performs best in Massachusetts. In Fig. 12, MSMDFF-Net^{DP} performs best in Boston. These visual results correspond to the quantitative experimental data in Table VIII. Furthermore,

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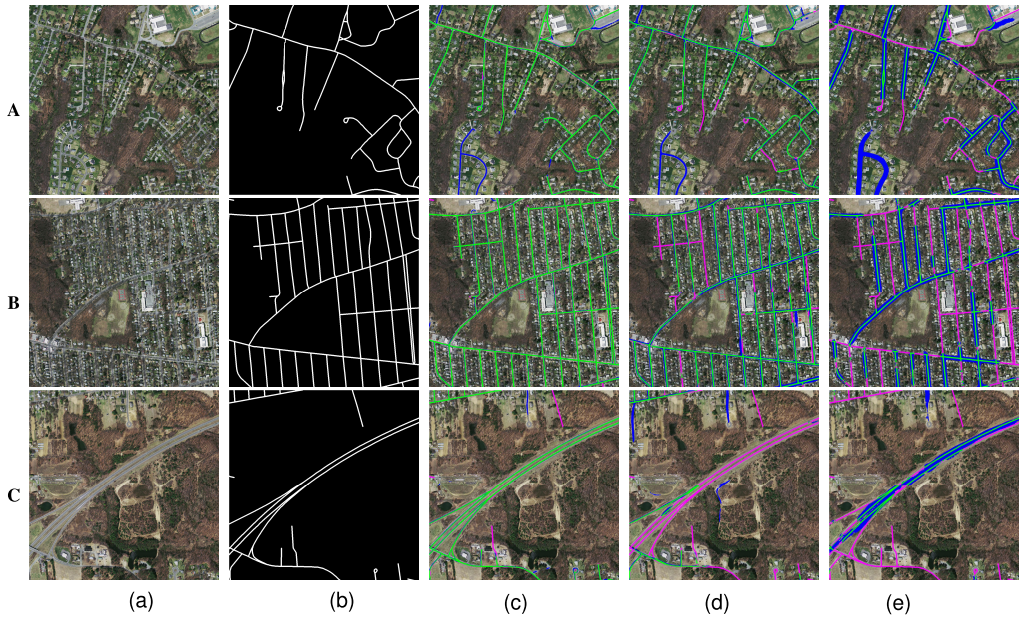


Fig. 11. Visual prediction results for three random areas in Massachusetts. Green color represents true positives, blue color represents false positives, and red color represents false negatives. (a) SRC. (b) LABEL. (c) MSMDFF-Net^{MA}. (d) MSMDFF-Net^{DP}. (e) MSMDFF-Net^{SP}.

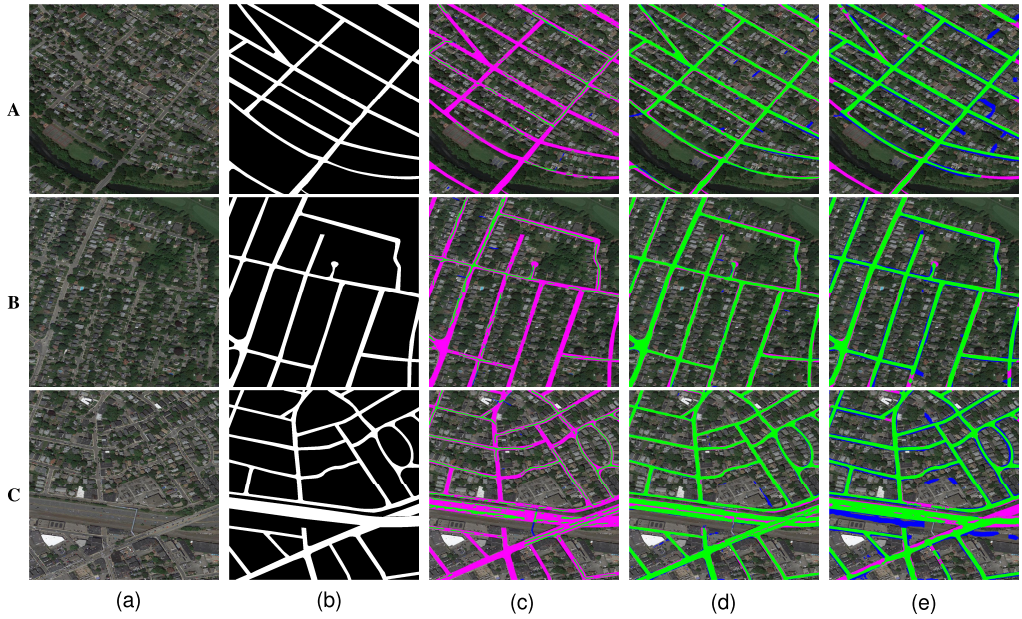


Fig. 12. Visual prediction results for three random areas in Boston. Green color represents true positives, blue color represents false positives, and red color represents false negatives. (a) SRC. (b) LABEL. (c) MSMDFF-Net^{MA}. (d) MSMDFF-Net^{DP}. (e) MSMDFF-Net^{SP}.

we can also observe that the resolution of the Massachusetts image is the same as the MA training dataset, while the resolution of the Boston image is closest to the DP training dataset. By considering the extraction results of the two models in these two regions, we can conclude that the impact of image resolution on road extraction results is related to the difference between the resolution of the training images and the resolution of the current experimental images. The smaller the resolution difference, the better the extraction results.

G. Ablation Study

In the subsequent experiments, "MA" is the Massachusetts dataset, "DP" stands for the DeepGlobe dataset, and "SP" identifies the SpaceNet dataset.

1) Effectiveness of Our Proposed Modules: In this article, in order to solve the problem of fragmented road extraction results, three modules MDFF, MSR, and MDCF are proposed. To evaluate the effectiveness of these the modules, the ablation experiment results on the three datasets are presented in Table IX. Comparing different combinations of our proposed modules, there is no universal optimal combination that can give the best extraction results on all datasets. Taking the number of highest and second-highest scores of the F1 and IoU in Table IX as the main reference, we give two suggestions for the using them.

1) MA dataset and SP dataset. The results of the combination MDFF+MSR+MDCF show that the three evaluation indicators R, F1, and IoU obtain the highest scores, so this configuration is optimal.

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TABLE IX

ABLATION STUDY RESULTS OF THE PROPOSED MDFF, MSR, AND MDCF ON THREE DATASETS (MA,DP, AND SP). IN THIS TABLE, SOME DATA ARE COLORED, WITH RED REPRESENTING THE HIGHEST VALUE OF THE CORRESPONDING EVALUATION METRIC AND BLUE INDICATING THE SECOND-HIGHEST VALUE

METHODs			P (%)			R (%)			F1 (%)			Road IOU (%)		
MDFF	MSR	MDCF	MA	DP	SP	MA	DP	SP	MA	DP	SP	MA	DP	SP
✓			**78.86**	77.61	**75.48**	64.3	**77.87**	71.84	70.84	77.74	73.61	54.84	63.59	58.25
		✓	**75.68**	77.01	73.24	65.51	75.68	73.55	70.23	76.34	73.39	54.11	61.73	57.97
✓	✓		74.7	77.72	72.16	68.28	**77.83**	**75.28**	71.35	77.78	73.69	55.46	63.63	58.34
✓		✓	75.19	**78.67**	**75**	**68.57**	77.69	74.59	**71.73**	**78.18**	**74.8**	**55.92**	**64.17**	**59.74**
✓	✓	✓	74.8	**78.92**	73.96	**69.14**	76.85	**76.03**	**71.86**	**77.87**	**74.98**	**56.07**	**63.76**	**59.98**

TABLE X

ABLATION RESULTS OF THE PROPOSED MDFF ON MA, DP, AND SP. "KERNEL_SIZE" IS THE SIZE OF THE STRIP CONVOLUTION KERNEL IN **MDFF**. THE DATA WITH RED COLOR REPRESENTS THE HIGHEST VALUE OF THE EVALUATION INDEX IN THAT COLUMN, WHILE THE BLUE DATA REPRESENTS THE SECOND-HIGHEST VALUE

MDFF	P (%)			R (%)			F1 (%)			Road IOU (%)		
	MA	DP	SP	MA	DP	SP	MA	DP	SP	MA	DP	SP
kernel_size = 3	77.78	78	73.85	65.85	**77.5**	**74.54**	71.32	**77.75**	**74.19**	55.42	**63.59**	**58.97**
kernel_size = 5	75.65	78.44	**74.42**	**68.4**	77.46	74.52	**71.85**	**77.95**	**74.47**	**56.06**	**63.87**	**59.32**
kernel_size = 7	76.83	**80.18**	72.22	**67.4**	71.35	74.51	71.81	75.51	73.34	**56.02**	60.65	57.91
kernel_size = 9	**78.86**	77.61	**75.48**	64.3	**77.87**	71.84	**70.84**	77.74	73.61	54.84	**63.59**	58.25
kernel_size = 11	**78.1**	**79.06**	71.84	66.11	74.64	**75.32**	71.61	76.79	73.54	55.77	62.32	58.15
kernel_size = 13	77	76.6	72.53	67.24	75.03	73.28	71.79	75.8	72.9	55.99	61.04	57.36

2) DP dataset. The combination of MDFF + MDCF is recommended. Experimental results show that both F1 and IoU obtain the highest score, P evaluation index is the second-highest, and the R value is 77.69%, which is almost equal to the second-highest value 77.83%.

2) Effect of different configurations for MDFF: MDFF-initblock contains four strip convolutions with different directions, which are used to enhance the feature extraction ability in the initial stage of the network. As shown in Table X, we analyze the performance of strip convolution with different parameters for different resolution images. When kernel size = 5, the MDFF module achieves the highest F1 and IoU scores on the DP and SP datasets. For the evaluation metric R, the difference between the current value and the highest value is very small, which are 0.05% (DP) and 0.03% (SP) smaller than the second-highest values, and 0.53% (DP) and 1.06% (SP) smaller than the highest values. So, it can be assumed that the kernel size = 5 for MDFF is the optimal configuration in DP and SP datasets. The MDFF also gets the best results on the MA dataset by using the same parameters as above. According to the previous analysis, when kernel size = 5, MDFF-initblock shows the best performance for three different resolution data. The MDFF module does not need to set different hyperparameters for RS images with different resolution, which will bring great convenience to practical applications.

3) Effect of different configurations for MDCF: Same as MDFF-initblock, MDCF-decoder also contains four strip convolutions of different directions. Different from MDFF, the SCM-D of MDCF are used in the recover stage of the feature map. In order to test the effectiveness of MDCF, the results on three different resolution data are shown in Table XI. On the MA dataset (1 m/pixel), when the size of the strip convolution kernel is set to 7, both F1 and IoU obtain the highest scores, and R obtains the second-highest score with 0.25% less than the highest score. So, the optimal parameter for strip convolution's convolution kernel is 7 on MA dataset (1 m/pixel). Similarly, the strip convolution kernel on the DP dataset (0.5 m/pixel) is set to 9 as the optimal parameter, while the strip convolution kernel on the SP dataset (0.3 m/pixel) is set to 11.

From the ablation results, it can be seen that setting the kernel size too large or too small would affect the model's accuracy, which can be explained by the receptive field of the convolution. In images with higher resolution, roads of the same width occupy more pixels, requiring larger kernel size. Using smaller kernel size may not cover the entire road width and can be easily affected by local similar features. For images with lower resolution, using larger kernel size can introduce distant and unrelated features. Therefore, the size of the SCM-D kernel in the MDCF-decoder is dependent on the resolution of the images used in the experiments. Based on the results of ablation experiments, we provide a guideline for selecting kernel size. For images with a resolution of 1 m/pixel, the kernel size is set to 7. For images with a resolution of 0.5 m/pixel, the kernel size is set to 9. For images with a resolution of 0.3 m/pixel, the kernel size is set to 11. Other resolution of images can refer to this approach for selecting kernel size.

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TABLE XI

ABLATION RESULTS OF THE PROPOSED MDCF ON MA, DP, AND SP. KERNEL_SIZE IS THE SIZE OF THE STRIP CONVOLUTION KERNEL IN **MDCF**. IN THIS TABLE, THE DATA WITH RED COLOR REPRESENTS THE HIGHEST VALUE OF THE EVALUATION INDEX IN THAT COLUMN, WHILE THE BLUE DATA REPRESENTS THE SECOND-HIGHEST VALUE

MDCF	P (%)			R (%)			F1 (%)			Road IOU (%)		
	MA	DP	SP	MA	DP	SP	MA	DP	SP	MA	DP	SP
kernel_size = 3	75.19	73.36	72.49	66.43	**77.41**	74.88	70.54	75.33	73.66	54.49	60.42	58.31
kernel_size = 5	**75.74**	76.82	71.99	65.96	74.16	**76.24**	70.51	**75.47**	**74.05**	54.45	**60.6**	**58.8**
kernel_size = 7	75.48	**77.44**	**73.22**	**66.68**	72.99	74.64	**70.81**	75.15	73.92	**54.81**	60.19	58.63
kernel_size = 9	**75.68**	77.01	**73.24**	65.51	**75.68**	73.55	70.23	**76.34**	73.39	54.11	**61.73**	57.97
kernel_size = 11	75.13	75.7	72.95	**66.85**	74.79	**75.38**	**70.75**	75.24	**74.15**	**54.74**	60.31	**58.91**
kernel_size = 13	75.18	**77.09**	72.52	66.53	73.36	74.68	70.59	75.18	73.58	54.55	60.23	58.21

TABLE XII

EXPERIMENTS WERE CONDUCTED ON A SINGLE NVIDIA TESLA P40, WITH ALL OTHER EXPERIMENTAL PARAMETERS CONSISTENT WITH THOSE LISTED IN TABLE II. THE MEAN TIMES OF TRAINING AND INFERENCE PRESENTED IN THIS TABLE REPRESENT THE TIME OF THE MODEL REQUIRES TO COMPUTE ONE IMAGE. THESE TIMES WERE CALCULATED BASED ON THE TRAINING AND INFERENCE TIMES FOR 100 BATCHES

Methods	P(%)	R(%)	F1(%)	Road IoU(%)	APLS(%)	Train time(s)	Inference time(s)
MSMDFF	78.92	76.85	77.87	63.76	70.15	0.1372	0.0431
MSMDFF#	77.87	77.63	77.75	63.59	70.04	0.1486	0.0456

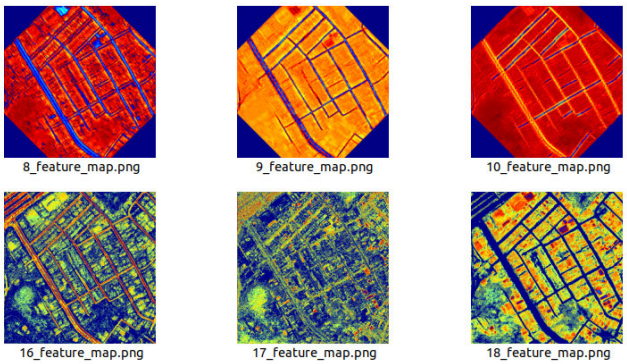


Fig. 13. Convolution output feature maps. The first row (8, 9, 10_feature_map.png) represents the feature maps obtained by using strip convolution(45°), and the second row (16, 17, 18_feature_map.png) represents the feature maps obtained by using square convolution.

V. DISCUSSION

A. Discussion of SCM-D

1) Comparison of SCM-D and SCM: To analyze the computational efficiency and accuracy of SCM-D and SCM, we replace the SCM-D with the SCM in the Decoder module of MSMDFF-Net, and named the network MSMDFF-Net#. We conduct experiments on the MSMDFF-Net and MSMDFF-Net# on the DeepGlobe dataset, as shown in Table XII.

From the experimental results, it is evident that there is a very small difference in the evaluation indicator scores between the MSMDFF-Net using the SCM-D module and the MSMDFF-Net# using the SCM module. Although in the experimental results, our proposed SCM-D does not have a very obvious advantage over the SCM proposed by CoANet in terms of computational efficiency and accuracy, our SCM-D holds several other significant advantages.

1) SCM-D exhibits superior computational efficiency. The SCM module as proposed by CoANet necessitates computations over a range that is twice as large as the feature maps, whereas SCM-D only requires computations within the original size range.
2) SCM-D can compute strip convolution in any direction, while SCM can only compute 45° direction strip convolution.
3) SCM-D can adeptly adapt to changes in the size of the feature map. For the SCM operation proposed by CoANet, when the stride of the convolution equals 2, the size of the feature map is halved, and the pixel correspondence is disappeared.

2) Visual Results of SCM-D: SCM-D is realized based on the rotation of feature map, which results in information loss at the corners during the computation of strip convolution in the 45° and 135° directions. To illustrate more intuitively that the method proposed in this article can automatically restore the lost information, the visualized computational results of the SCM-D of the decoder are presented, as depicted in Fig. 13.

From the visualization results, it can be seen that the feature maps obtained from square convolution have no information loss, while the computation results of the 45° convolution lose information at the corners. When constructing the MDFF-initblock (Fig. 4) and MDCF-decoder (Fig. 7) using SCM-D, we combine four-directional strip convolutions and square convolution. Some of the fusion results in the MDCF-decoder are shown in Fig. 14. By using square convolution in conjunction with the SCM-D module, it is clear that the information loss at the corners is effectively addressed.

At the same time, we can also observe the distinct roles of strip convolution and square convolution. Strip convolution is more sensitive to line features, with the long-distance linear receptive field playing a crucial role in solving the fragmentation of road extraction results. For instance, the feature maps obtained through strip convolution exhibit more pronounced linear features of roads, and the distribution of background pixels is relatively uniform. Conversely, the feature maps obtained by using square convolution present a variety of background pixel types and similar road textures, which can potentially impact the accuracy of road generation. And the fused feature maps (Fig. 14) render the background pixels

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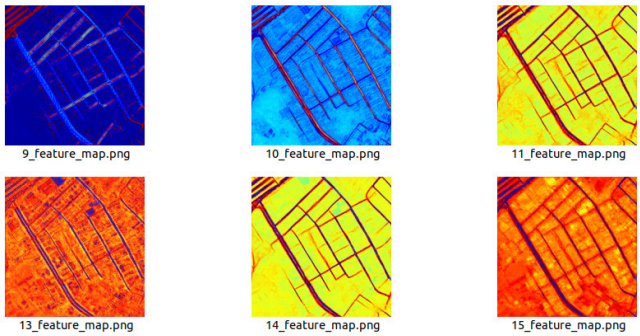


Fig. 14. Fused convolution output feature maps.

TABLE XIII

RUNNING TIMES OF OUR PROPOSED MSMDFF-NET AND SEVERAL OTHER STATE-OF-THE-ART ROAD EXTRACTION METHODS ARE COMPARED. THE AVERAGE TRAINING AND TESTING TIMES FOR A SINGLE IMAGE ARE LISTED, BASED ON 100 BATCHES OF TRAINING AND INFERENCE TIMES

Index	Methods	Training(s)	Inference(s)
A	UNet++	0.3954[10]	0.1186[11]
B	DeepRoadMapper	0.0873[5]	0.0265[5]
C	LinkNet34	0.0358[2]	0.0151[2]
D	LinkNet50	0.0685[4]	0.0235[4]
E	DlinkNet34	0.0494[3]	0.0158[3]
F	DlinkNet50	0.3464[9]	0.0390[7]
G	RoadCNN	0.0944[6]	0.0330[6]
H	GAMSNet	0.0290[1]	0.0095[1]
I	CoANet	1.1431[12]	0.0986[10]
J	CADUNet	0.1566[8]	0.0475[9]
K	SGCNet	0.6310[11]	0.1601[12]
L	MSMDFF-Net	0.1374[7]	0.0430[8]

more uniform, thereby facilitating the model in distinguishing the subject (road) from the background (object).

B. Computational Efficiency

As shown in Table XIII, we count the running time of MSMDFF-Net and several other state-of-the-art road extraction methods on the MA. The input data size of the experiments for all methods is 512 × 512, data type is flaot32, batch size is set to 4, and they are all performed on the 1 × NVIDIA TESLA P40. In the training stage, we only count the time from the forward calculation to the gradient backpropagation. In the inference stage, we only count the time of the forward calculation. In Table XIII, we calculate the average time for one image based on the training and inference times for 100 batches.

It can be noticed that the training time and inference time of our MSMDFF-Net are not the fastest. These operations, such as downsampling and feature map rotation, require a large computational cost, which affects the training efficiency and inference efficiency of our method. In the future, we will consider placing the downsampling operation into the parallel pipeline dataloader for reducing the time cost of the model.

VI. CONCLUSION

In this article, we proposed a novel network, MSMDFF-Net, to reduce the fragmentation of road extraction results from RS images. Within this network, a new method called SCM-D was developed for calculating strip convolutions in arbitrary directions. By using SCM-D, MDFF-initblock and MDCF-decoder were built. The use of SCM-D can increase the linear receptive field of the model and learn the long context relationship between neighboring pixels, which are very useful to improve the integrity of road. Besides, MSR-encoder was proposed for improving the generalization of the model on different resolution RS image, which can make full use of the multiscale information of different coding stages. Moreover, we put in a lot of effort to build a pytorch-based road extraction platform, which permits different road extraction models to share the same experimental parameters, leading to more scientific and reliable comparisons. Extensive experiments on popular benchmarks (Massachusetts, DeepGlobe, SpaceNet datasets) at different resolution demonstrated that our proposed MSMDFF-Net outperforms several other state-of-the-art methods. Ablation experiments were also conducted to showcase the effectiveness and superiority of MDFF, MSR, and MDCF modules. However, both the training time and inference time of MSMDFF-Net are not the fastest. Future research will focus on accelerating the algorithm's calculation speed. Given the excellent performance of strip-shaped convolution in our experiments, more complex cross-directional strip-shaped convolution may yield superior results. We intend to delve deeper into this aspect in our future work.

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